

**DINESHKUMAR LOGANATHAN**

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**Academic Details:**

| <b>Name of the Degree</b>       | <b>Course</b>                               | <b>Name of the College</b>   | <b>Name of the University</b>                                               | <b>Grade</b>                  | <b>Year of Passing</b> |
|---------------------------------|---------------------------------------------|------------------------------|-----------------------------------------------------------------------------|-------------------------------|------------------------|
| Doctor of Philosophy (Ph.D)     | Mechanical Engineering                      | College of Engineering       | National Cheng Kung University (NCKU), Tainan, Taiwan. (QS World Rank: 191) | CGPA-9.4/10                   | 2026                   |
| Master of Technology (M.Tech)   | Engineering Design (Mechanical Engineering) | Amrita School of Engineering | Amrita Vishwa Vidyapeetham, Coimbatore, India                               | CGPA- 9.6/10 (Gold Medalists) | 2016                   |
| Bachelor of Technology (B.Tech) | Mechanical Engineering                      | Amrita School of Engineering | Amrita Vishwa Vidyapeetham, Coimbatore, India                               | CGPA-7.41/10                  | 2014                   |

**Experience details:**

| <b>Institution Name</b>                                                                                    | <b>Designation</b>                                        | <b>Years of experience</b> |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------|
| Indian Institute of Technology (IIT) - Jammu, India                                                        | Junior Research Fellow                                    | Jan-2020 to Sep-2021       |
| Amrita School of Engineering, Bangalore, India                                                             | Assistant Professor, Department of Mechanical Engineering | July-2016 to Dec-2019      |
| Universitat Politècnica de Catalunya, CTTC (Heat and Mass Transfer Technological center), Barcelona, Spain | Research fellow come Master Student.                      | Feb-2016 to June-2016      |



**Major academic achievements:**

1. Received the **2025 CTCI Foundation Science and Technology Scholarship** (Research Scholarship for Overseas Students) during the Ph.D. program at National Cheng Kung University (NCKU), Tainan, Taiwan.
2. Received **Veritas et Conscientia Scholarship (VCS)** during my period 2024-2025 and 2021-2022 of the Ph.D. program at National Cheng Kung University (NCKU), Tainan, Taiwan.
3. Received the '**Outstanding Research Scholar**' award from National Cheng Kung University during the academic year 2024-2025.
4. Received the '**Excellent Research Scholar**' award from National Cheng Kung University during the academic year 2023-2024.
5. Participated in **Stanford × Taiwan Biodesign Program and Medical Equipment Product Design Talent Training Course**, and secured **Fourth** place in the overall competition held at the end of the semester on 15<sup>th</sup> July 2023.
6. Successfully completed the **15 credits of Innovative Biodesign and Intellisoftware Development (IBID) Program** during September 2022 to June 2023 from the department of Biomedical Engineering at NCKU, sponsored by **MOE, Taiwan**.
7. The article titled 'Untethered Magnetic Microswimmers for Targeted Particle Transport and Flow Manipulation' was chosen as the **front cover image** of the journal '*Advanced Materials Technologies*' (Volume 11, Issue 6, e70823, 2026)
8. The article titled 'Behavioural responses of zebrafish with sound stimuli in microfluidics' was chosen as the **back cover image** of the journal '*Lab on a Chip*' (Issue 1, 2023)
9. The article titled 'An On-Demand Microrobot with Building Block Design for Flow Manipulation' was chosen as the **back cover image** of the journal '*Advanced Materials Technologies*' (Issue 3, Volume 8)
10. Presented the research topic titled 'Acoustically assisted spatial habituation of zebrafish larvae in microfluidics' at the 2022 Taiwan Zebrafish Symposium organized by the 'Institute of Molecular and Genomic Medicine, NHRI' on December 3, 2022.

**Research papers published:**

**International Journals:**

1. **Dineshkumar Loganathan** and Chia-Yuan Chen\*, 'An Intelligent Microfluidics Enabled by Data-Driven Deep Learning-Based Microrobot-Assisted Flow Manipulation,' *Advanced Intelligent Systems*, 2026 (Accepted on the 7<sup>th</sup> of July 2026) (IF: 6.7 Rank: 15/88, Q1).
2. **Dineshkumar Loganathan** and Chia-Yuan Chen\*, 'Reconfigurable Cilia-Based Magnetic Millirobots for Cooperative Particle Manipulation Through Programmable Assembly in Microfluidics,' *Micromachines*, 2026, 17 (7), 834. (IF: 3.5 Rank: 30/81, Q2).



3. **Dineshkumar Loganathan**, Pu-Hsiang Wang, Yueh-Hsun Lu, and Chia-Yuan Chen\*, 'Programmable microfluidics for zebrafish larvae manipulation using multiple magnetic microrobots,' *Sensors and Actuators Reports*, 2026, 12, 100484. (IF: 6.9 Rank: 6/81, Q1).
4. **Dineshkumar Loganathan**, Dong-Ming He, Kai-Wei Chen, and Chia-Yuan Chen\*, 'Untethered Magnetic Microswimmers for Targeted Particle Transport and Flow Manipulation,' *Advanced Materials Technologies*, 2025, 11 (6), e70823. (IF: 6.2, Rank: 122/461, Q2).
5. **Dineshkumar Loganathan**, Tung Ou Yang, Chia-Yun Chen\*, Chia-Yuan Chen\*, 'Magnetic Cilia with Programmable Beating Patterns for Vortex-Driven Mixing in Microfluidics,' *Langmuir*, 2025, 41, 21562-21575. (IF: 3.9, Rank: 88/239, Q2).
6. Prashant Kishor Sharma, Po-Wei Wei, **Dineshkumar Loganathan**, Yueh-Hsun Lu, ChiaYuan Chen\*, 'Microflow Switching Using Artificial Cilia for On-Demand Particle Manipulation,' *Advanced Intelligent Systems*, 2025, 2500431. (IF: 6.1, Rank: 7/48, Q1).
7. Prashant Kishor Sharma, **Dineshkumar Loganathan**, Ming-Lung Chen, Yueh-Hsun Lu, Pu-Hsiang Wang, Chia-Yuan Chen\*, 'Cognitive dynamics of drug-mediated zebrafish under sound stimuli in a microfluidic environment,' *Biomicrofluidics*, 2025, 19, 034105. (IF: 2.6, Rank: 12/40, Q2).
8. **Dineshkumar Loganathan**, Chia-Hsin Cheng, Po-Wei Wei, Chia-Yuan Chen\*, 'In-Liquid Micromanipulation via a Magnetic Microactuator for Multitasking,' *Small Science*, 2025, 5, 2500010. (IF: 11.2, Rank: 43/439, Q1).
9. Kai-Hsiang Yang, **Dineshkumar Loganathan**, Ming-Lung Chen, Vignesh Sahadevan, Chia-Yun Chen\*, Chia-Yuan Chen\*, 'Enhancement of zebrafish sperm activation through microfluidic mixing induced by aquatic microrobots,' *Microfluidics and Nanofluidics*, 2025, 29, 1. (IF: 2.5, Rank: 13/41, Q2).
10. **Dineshkumar Loganathan**, Lin-Tuz-En Kao, Chih-Hsuan Cheng, Chia-Yuan Chen\*, 'Hemodynamic analysis of mean arterial pressure on cerebral arteries for embolic ischemic stroke using a lumped parameter model,' *IEEE Access*, 2024, 12, 197942. (IF: 3.4, Rank: 87/250, Q2).
11. **Dineshkumar Loganathan**, Chen-Yi Ou, Chao-Wei Hsu, Chia-Yuan Chen\*, 'A Control Strategy of Multiple Microrobots Using a Hybrid Electromagnetic System,' *Advanced Materials Technologies*, 2024, 9, 2401135. (IF: 6.8, Rank: 62/345, Q1).
12. **Dineshkumar Loganathan**, Chia-Ling Hsieh, Chen-Yi Ou, and Chia-Yuan Chen\*, 'A stepwise control of multiple millirobots for flow manipulation applications,' *Advanced Intelligent Systems*, 2024, 7, 2300483. (IF: 7.4, Rank: 62/345, Q1).
13. **Dineshkumar Loganathan**, Ying Chaung, Yueh-Hsun Lu, Chia-Hsin Cheng, and ChiaYuan Chen\*, 'An untethered magnetic microgripper for high-throughput micromanipulation,' *Advanced Materials Technologies*, 2024, 9, 2400292. (IF: 6.8, Rank: 62/345, Q1).
14. **Dineshkumar Loganathan**, Chia-Ling Hsieh, Bo-Er Shi, Yueh-Hsun Lu, and Chia-Yuan Chen\*, 'An On-Demand Microrobot with Building Block Design for Flow Manipulation,' *Advanced Materials Technologies*, 2023, 8, 2201073. (Impact factor: 8.856, Rank: 62/345, Q1).
15. **Dineshkumar Loganathan**, Shu-Heng Wu, and Chia-Yuan Chen\*, 'Behavioural responses of zebrafish with sound stimuli in microfluidic,' *Lab on a Chip*, 2023, 23, 106-114. (Impact factor: 7.517, Rank: 7/84, Q1).
16. Vignesh Sahadevan, **Dineshkumar Loganathan**, Ying Chuang, Chien Feng Lo, Chia-Yun Chen\*, and Chia-Yuan Chen\*, 'Synergetic benefits of microfluidics using artificial cilia and ZnO/SnFe<sub>2</sub>O<sub>4</sub> for the degradation of pollutants,' *Materials Chemistry and Physics*, 2023, 307, 128068. (Impact factor: 4.778, Rank: 125/345, Q2).
17. Satishkumar Subendran, Chun-Fang Wang, **Dineshkumar Loganathan**, Yueh-Hsun Lu & Chia-Yuan Chen\*, 'An aquatic microrobot for microscale flow manipulation,' *Scientific Reports*, 2022, 12, 5041. (Impact factor: 4.997, Rank: 19/135, Q1).
18. N. Mohammed, Jithin, E. V., **Dineshkumar, L.**, V. Kishore, R. \*, and Mohammad, A. \*, "Tip Opening



of Burner-Stabilized Flames”, *Energy and Fuels*, vol. 32, pp. 2344-2354, **2018**. Impact factor- SCI (3.5).

#### **International Conferences:**

19. **Dineshkumar Loganathan**, Tung Ou Yan, Chia-Yuan Chen\*. ‘Hydrodynamic Control via Programmable Magnetic Cilia for Enhanced Microfluidic Mixing in Photodegradation,’ *2026 IEEE 39th International Conference on Micro Electro Mechanical Systems (MEMS)*, Salzburg, Austria, 2026, pp. 1536-1539,
20. Hydrodynamic Control via Programmable Magnetic Cilia for Enhanced Microfluidic Mixing in Photodegradation. *2026 IEEE 39th International Conference on Micro Electro Mechanical Systems (MEMS)*. 1536-1539, 2026.
21. Siddhartha Gollamudi and **Dineshkumar L\***. ‘Numerical analysis on enhancement of recirculation in porous medium to improve heat transfer rate.’ *AIP Conf. Proc.* 2 November 2020; 2273 (1): 050069.
22. Sanjana Reddy D, Y Ranjani, Mukeshanand M, M Sudharshan Srinivas, D. Lalith Kumar, and **Dineshkumar L\***. ‘Characteristics of mixed convective flow over fins for automobile application.’ In *Proceedings of the Emerging Trends in Mechanical Engineering*, Singapore, 2020//, 2020; pp. 105–113.
23. K. Aditya Sai, S. Manikanta, P. Vignesh, Naveen Kumar, P. Sai Teja, **L. Dineshkumar\***, and V. Vignesh. ‘A review on preparation, efficiency, and emissions of Mahua biodiesel and its blends.’ *AIP Conf. Proc.* 20 December 2019; 2200 (1): 020093.
24. Arjun Damodar E.T and **Dineshkumar L\***. ‘Numerical Analysis on Enhancement of Heat Transfer from a Heated Surface (Fins).’ *AIP Conf. Proc.* 20 December 2019; 2200 (1): 020063.
25. Gagan Chakravarthy C G, Aswin Krithik, G Siddhartha, and **Dineshkumar L\***. ‘Numerical study on stabilisation of flame inside micro-combustor.’ *AIP Conf. Proc.* 20 December 2019; 2200 (1): 020058.
26. V S GopalaKrishna and **Dineshkumar L\***. ‘Numerical analysis on enhancement of heat transfer in micro-heat sink using dimpling surface.’ *AIP Conf. Proc.* 20 December 2019; 2200 (1): 020059.
27. Adsul, Pranita & **Dineshkumar L\***. ‘On code verification of 2D transient heat conduction in composite wall.’ *2018 IOP Conf. Ser.: Mater. Sci. Eng.* 377 012128.
28. Kumar Doreti, Lalith & **Dineshkumar L\***. ‘Control techniques in flow past a cylinder-A Review.’ *2018 IOP Conf. Ser.: Mater. Sci. Eng.* 377 012144.
29. A. S. Divakar Shetty\*, **L. Dineshkumar**, Sandeep Koundinya, and Swetha K. Mane. ‘Experimental investigation on CRDI engine using butanol-biodiesel-diesel blends as fuel.’ *AIP Conf. Proc.* 19 July 2017; 1859 (1): 020028.

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**Journal Reviewer:**

1. Nature Communications (IF: 15.7) in 2026 (2 times)
2. Scientific Reports (IF: 3.9) in 2025 and 2026
3. Journal of Micro and Bio Robotics (IF: 1.1) in 2026
4. BioNanoScience (IF: 3.2) in 2025
5. npj Robotics in 2025

**International Research Experience:**

| Institution Name                                                                                                 | Designation                             | Years of experience   |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------|
| Universitat Politècnica de Catalunya,<br>CTTC (Heat and Mass Transfer<br>Technological centre), Barcelona, Spain | Research fellow come<br>Master Student. | Feb-2016 to June 2016 |

**International Research Collaboration:**

| Institution Name                                         | Research Centre Name                                                | Remarks                                                          |
|----------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------|
| King Abdulaziz University, Jeddah<br>21589, Saudi Arabia | King Abdulaziz university's<br>High-Performance<br>Computing Centre | Have employed<br>Aziz<br>Supercomputer for<br>computational work |

**Funded Research Projects:**

1. As a Co-PI - Titled 'Hybrid solar-based smart dryer for copra production', funded by the Ministry of Food Processing Industries (MoFPI) for a project duration of 2 years, with a funding amount of 16,70,000 (INR).
2. As a Principal Investigator – Titled ‘Design and fabrication of micro-combustor as the replacement of chemical battery in mobile phones’ funded by ‘Dassault Systèmes Foundation’ for a project duration of 2 years, with a funding amount of 2,50,000 (INR).

**Areas of Interest:**

- Fluid Mechanics.
- Computational fluid dynamics (CFD).

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- Finite Element Analysis.
- Internal combustion engine.

**Technical software known:**

- ▯ Ansys, Abaqus, Fluent, MATLAB, Solid Edge/Work, and C++

**Co-Curricular Achievements:**

- **Class Representative** for XI and XII.
- Secured Second Rank from the academic year 2007 to 2010.

**Industrial Visits/ In-Plant Training:**

- In-Plant training on **Internal Combustion Engine** at TNSTC.. Tirupur (2012).
- In-Plant training on **Wheels and services** at ABT Mahalingam Service, Palladam (2013).
- In-Plant training on **Paper Manufacturing Technology** in TNPL, Karur (2012).

**Extra-Curricular Activities:**

- Secured first place in the **Inter Dojo Sports Karate Championship** (2007).
- Got **Runner Up** in Zonal Level Volleyball match (2010).
- School people leader (SPL) from the academic year 2006 to 2008.
- Participated in **AMALA BHARATHAM CAMPAIGN (ABC)** during the academic year 2013-2014.

**Hobbies:**

- ▯ Cooking.
- ▯ Playing volleyball.
- ▯ Listening to music.
- ▯ Singing.

**Personal Details:**

Date of Birth: 27<sup>th</sup> January 1993.

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Languages Known: English and Tamil.

Father's Name: Mr. Loganathan L.

Mother's Name: Mrs. Mahalakshmi L.

Permanent Address: 21/A, Chinneya Gounder Street, P.Vadugapalayam, Palladam, Tirupur

Place: Tainan, Taiwan.

04/01/2026

(DINESHKUMAR LOGNATHAN)